

Avinash Malladi

malladiavinash3@gmail.com | (+91) 6281855895 | Portfolio | LinkedIn | GitHub | Credly

Summary

B.Tech Computer Science student with practical knowledge in software development, backend systems, and RESTful APIs using TypeScript, Node.js, Express, Python, PostgreSQL, and Redis. Experienced in distributed task queues with BullMQ, Docker containerization, and Elasticsearch, with a solid foundation in Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), and problem solving. Interested in building scalable backend systems for e-commerce and SaaS platforms.

Education

SR University , Warangal <i>B.Tech in Computer Science and Engineering</i>	Aug 2023 – Present <i>CGPA: 8.6 / 10</i>
Narayana Junior College , Hyderabad <i>Intermediate (11th & 12th)</i>	Aug 2021 – May 2023 <i>Score: 92.9%</i>
Krishnaveni Talent School , Mancherla <i>High School (10th Standard)</i>	2020 <i>CGPA: 10 / 10</i>

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, C++, C, SQL

Backend & Systems: Node.js, Express.js, REST APIs, BullMQ, Redis, Rate Limiting, Asynchronous Processing

Databases & Search: PostgreSQL, MySQL, SQLite, Elasticsearch, Redis Caching

Core Computer Science: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design, DBMS, OS

Cloud & DevOps: AWS, Docker, Git, GitHub Actions, CI/CD, Postman, Linux, Render, Railway

Software Engineering: Agile, Scrum, Teamwork, Problem Solving, Code Reviews, Communication

Web Technologies: React.js, Tailwind CSS, HTML, CSS, Google OAuth 2.0, Webhooks, Slack API

Machine Learning & AI: RAG, LangChain, BM25, Scikit-learn, XGBoost, SHAP, LIME

Projects

ReachInbox – Distributed Asynchronous Job Scheduler | *TypeScript, BullMQ, Redis, PostgreSQL, ES*

- Designed an email scheduling service in TypeScript, Node.js, and Express, accepting CSV/TXT batch uploads up to 5k recipients and queuing delayed background jobs using BullMQ and Redis.
- Implemented multi-instance rate limiting with Redis sliding-window counters (capping 50 emails/hr per sender, 200 global limit), sending Slack alerts and rolling burst overflows into subsequent hours to achieve 100% execution reliability without dropped jobs.
- Integrated full-text search with Elasticsearch (falling back to PostgreSQL ILIKE) and user authentication with Google OAuth 2.0; containerized the stack with Docker.

SR University AI Assistant – Agentic RAG Chatbot | *Python, LangChain, BM25, Docker, CI/CD*

- Collaborated with academic peers to build an agentic RAG query assistant for university guidelines, achieving page-level citations across a benchmark of 150+ queries.
- Built a custom BM25 retrieval pipeline with document routing, chunking, and reranking, which improved retrieval hit rate to 89% and MRR to 0.82.
- Created agentic tool workflows for handbook search, calculations, and web search with automated fallbacks; containerized using Docker.

Smart Disaster Relief System | *Python, Flask, SQLite, Leaflet.js, OpenStreetMap*

- Engineered a Flask and SQLite web application coordinating in an Agile team to manage disaster reports, relief camps, resources, and victim allocation across 20+ camps.
- Optimized emergency dispatch via OpenStreetMap Nominatim and rule-based nearest-camp detection, which reduced response time by 35%.
- Constructed a real-time monitoring dashboard with JavaScript, Leaflet.js, and Chart.js, featuring periodic polling and threshold alerts for high-risk zones.

Explainable AI-Powered Lung Cancer Prediction | *XGBoost, SHAP, LIME, IEEE ICICCS 2026*

[Verify]

- Co-authored and presented a peer-reviewed research paper at the IEEE ICICCS 2026 conference on explainable machine learning for cancer prediction.
- Evaluated multiple ML algorithms, achieving 98% prediction accuracy using XGBoost; implemented SHAP and LIME for feature interpretation.

Certifications

Amazon Certified Cloud Practitioner – Amazon Skill Builder

May 2026 [Verify]

Microsoft Azure AI Fundamentals (AI-900)

Aug 2025 [Verify]

AWS Academy Data Engineering – AWS

Feb 2025 [Verify]

Python Full Stack – EduSkills

Dec 2024 [Verify]